



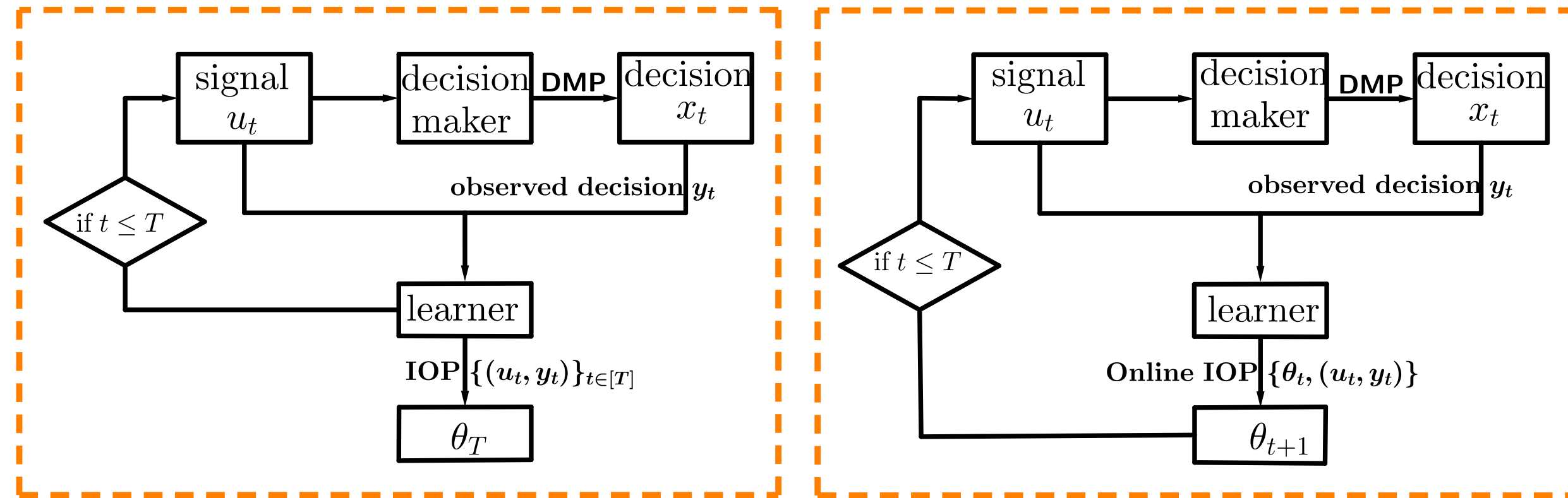
# Generalized Inverse Optimization through Online Learning

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## Introduction

- To unlock the potential of inverse optimization, we aim to elicit decision maker's preferences or restrictions through online learning in this paper.
- We develop an online learning algorithm to derive unknown parameter occurring in either the objective function or constraints.
- We also show that the online learning algorithm has a  $\mathcal{O}(\sqrt{T})$  regret bound.



## Decision making problem

We consider a family of parameterized optimization problem, in which  $\mathbf{x} \in \mathbb{R}^n$  is the decision variable,  $u \in \mathbb{R}^m$  is the external signal, and  $\theta \in \mathbb{R}^p$  is the parameter.

$$\begin{aligned} \min_{\mathbf{x} \in \mathbb{R}^n} \quad & f(\mathbf{x}, u, \theta) \\ \text{s.t.} \quad & \mathbf{g}(\mathbf{x}, u, \theta) \leq \mathbf{0} \end{aligned} \quad \text{DMP}$$

where  $f: \mathbb{R}^n \times \mathbb{R}^m \times \mathbb{R}^p \mapsto \mathbb{R}$  is a real-valued function, and  $\mathbf{g}: \mathbb{R}^n \times \mathbb{R}^m \times \mathbb{R}^p \mapsto \mathbb{R}^q$  is a vector-valued function. We denote  $X(u, \theta) = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{g}(\mathbf{x}, u, \theta) \leq \mathbf{0}\}$  the feasible region of DMP. We let  $S(u, \theta) = \arg \min\{f(\mathbf{x}, u, \theta) : \mathbf{x} \in X(u, \theta)\}$  be the optimal solution set of DMP.

## Inverse optimization and online setting

### Inverse optimization

Specifically, we consider a situation where a decision  $y_i$  (with respect to an input signal  $u_i$ ) for  $i \in [N]$  are observed and recorded, the IOP model can be formulated to minimize an empirical loss as in the following

$$\min_{\theta \in \Theta} \frac{1}{N} \sum_{i=1}^N l_\theta(u_i, y_i) \quad (\text{Batch-IOP})$$

where  $\theta$  denotes the parameter in DMP, and  $l_\theta(u_i, y_i)$  is a loss function that captures the discrepancy between the model inferred from data and the actual one.

### Online setting

- In our online learning setting, the signal-noisy decision pair become available to the learner one by one. Hence, the learning algorithm produces  $(\theta_1, \dots, \theta_{T+1})$ .
- Let  $l(\mathbf{y}_t, u_t, \theta_t)$  denote the loss the learning algorithm suffers when it tries to predict the  $t$ th decision given  $u_t$  based on  $\{(u_1, \mathbf{y}_1), \dots, (u_{t-1}, \mathbf{y}_{t-1})\}$ .
- The goal of the learner is to minimize the regret, which is the cumulative loss  $\sum_{t \in [T]} l(\mathbf{y}_t, u_t, \theta_t)$  against the possible loss when the whole batch of decisions are available. Formally, the regret is defined as

$$R_T = \sum_{t \in [T]} l(\mathbf{y}_t, u_t, \theta_t) - \min_{\theta \in \Theta} \sum_{t \in [T]} l(\mathbf{y}_t, u_t, \theta).$$

## Online learning algorithm

- Given a (signal, noisy decision) pair  $(u, \mathbf{y})$  and a hypothesis  $\theta$ , we set the loss function as the minimum (squared) distance between  $\mathbf{y}$  and the optimal solution set  $S(u, \theta)$  in the following.

$$l(\mathbf{y}, u, \theta) = \min_{\mathbf{x} \in S(u, \theta)} \|\mathbf{y} - \mathbf{x}\|_2^2. \quad \text{Loss Function}$$

- Once receiving the  $t$ th (signal, noisy decision) pair  $(u_t, \mathbf{y}_t)$ ,  $\theta_{t+1}$  can be obtained by solving the following optimization problem:

$$\theta_{t+1} = \arg \min_{\theta \in \Theta} \frac{1}{2} \|\theta - \theta_t\|_2^2 + \eta_t l(\mathbf{y}_t, u_t, \theta), \quad (1)$$

where  $\eta_t$  is the learning rate in round  $t$ , and  $l(\mathbf{y}_t, u_t, \theta)$  is defined in (Loss Function).

### Algorithm 1 Implicit Online Learning for Generalized Inverse Optimization

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1: Input: (signal, noisy decision) pairs  $\{(u_t, \mathbf{y}_t)\}_{t \in [T]}$ 
2: Initialization:  $\theta_1$  could be an arbitrary hypothesis of the parameter.
3: for  $t = 1$  to  $T$  do
4:   receive  $(u_t, \mathbf{y}_t)$ 
5:   suffer loss  $l(\mathbf{y}_t, u_t, \theta_t)$ 
6:   if  $l(\mathbf{y}_t, u_t, \theta_t) = 0$  then
7:      $\theta_{t+1} \leftarrow \theta_t$ 
8:   else
9:     set learning rate  $\eta_t \propto 1/\sqrt{t}$ 
10:    update  $\theta_{t+1} = \arg \min_{\theta \in \Theta} \frac{1}{2} \|\theta - \theta_t\|_2^2 + \eta_t l(\mathbf{y}_t, u_t, \theta)$  (solve (1))
11:   end if
12: end for

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## Theoretical analysis

- Regret bound

**Theorem 1.** Under suitable conditions, choosing  $\eta_t = \frac{D\lambda}{2\sqrt{2}(B+R)\kappa\sqrt{t}}$ , we have

$$R_T \leq \frac{4\sqrt{2}(B+R)D\kappa}{\lambda} \sqrt{T}.$$

where  $\lambda$  and  $\kappa$  are related to the smoothness of the objective functions.

- Consistency of the estimator

**Theorem 2.** Under suitable conditions, choosing  $\eta_t = \frac{D\lambda}{2\sqrt{2}(B+R)\kappa\sqrt{t}}$ , we have

$$\frac{1}{T} \sum_{t \in [T]} l(\mathbf{y}_t, u_t, \theta_t) \xrightarrow{p} \mathbb{E} l(\mathbf{y}, u, \theta^*)$$

where  $\theta^* \in \arg \min_{\theta \in \Theta} \mathbb{E} l(\mathbf{y}, u, \theta)$  which minimizes the true risk.

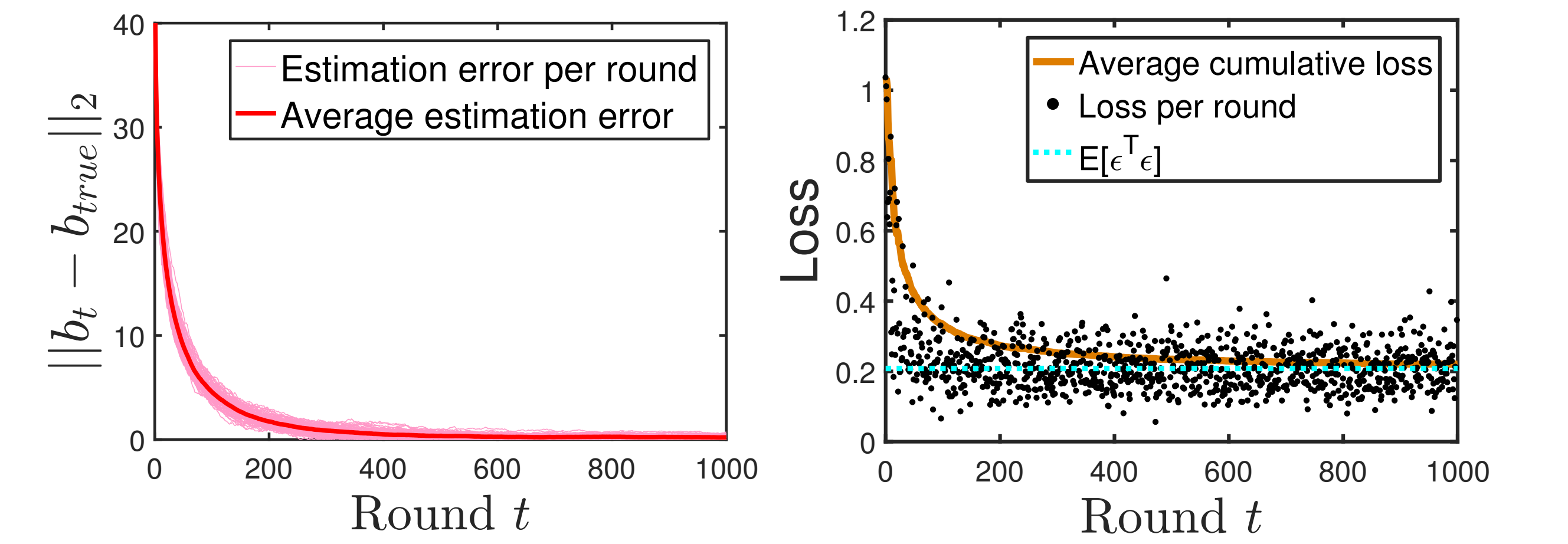
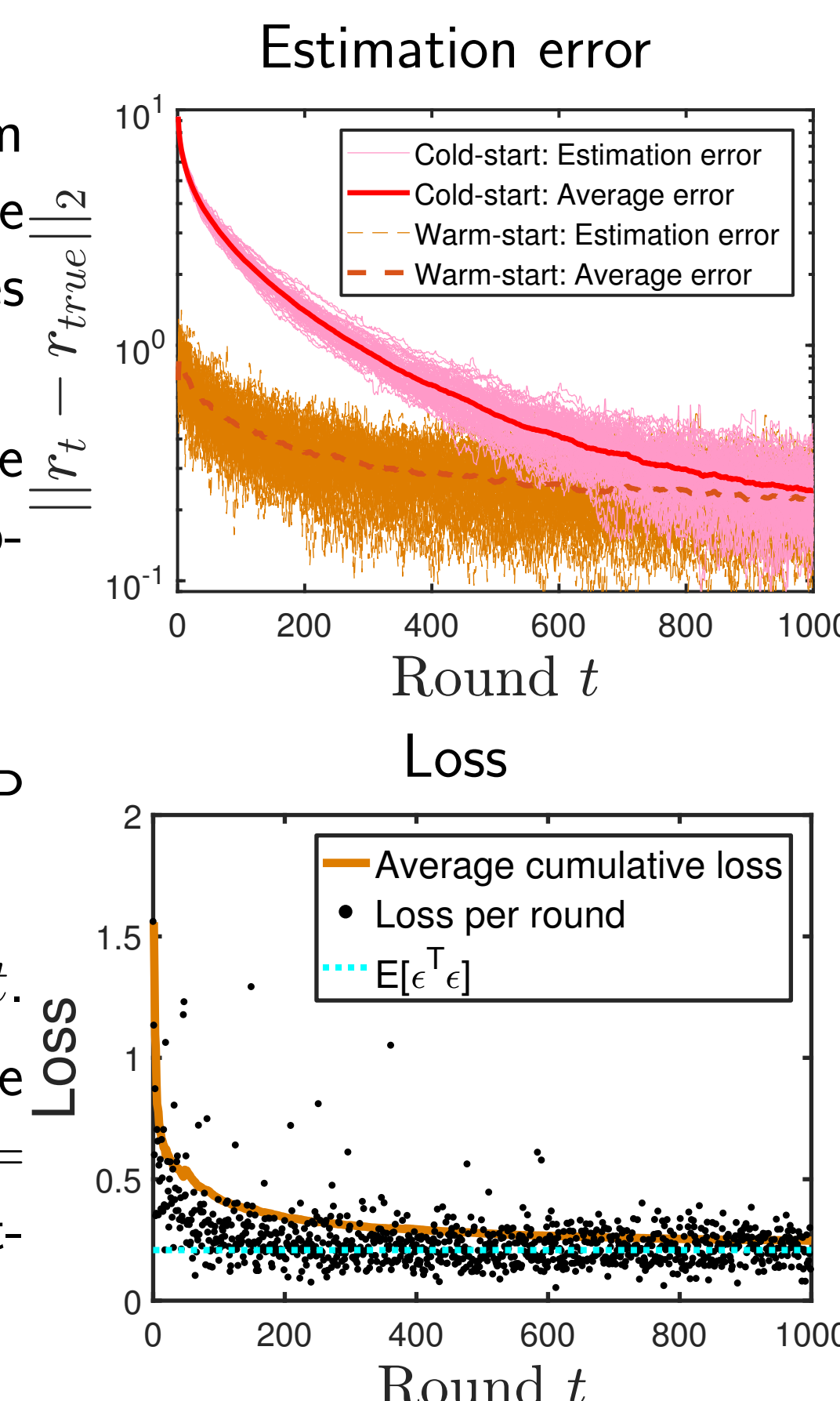
## Learning consumer behavior

- We now study the consumer's behavior problem in a market with  $n$  products. The prices for the products are denoted by  $\mathbf{p}_t \in \mathbb{R}_+^n$  which varies over different time  $t \in [T]$ .
- The consumer's decision making problem can be stated as the following utility maximization problem (UMP)

$$\begin{aligned} \max_{\mathbf{x} \in \mathbb{R}_+^n} \quad & u(\mathbf{x}) \\ \text{s.t.} \quad & \mathbf{p}_t^T \mathbf{x} \leq b \end{aligned} \quad \text{UMP}$$

where  $\mathbf{p}_t^T \mathbf{x} \leq b$  is the budget constraint at time  $t$ .

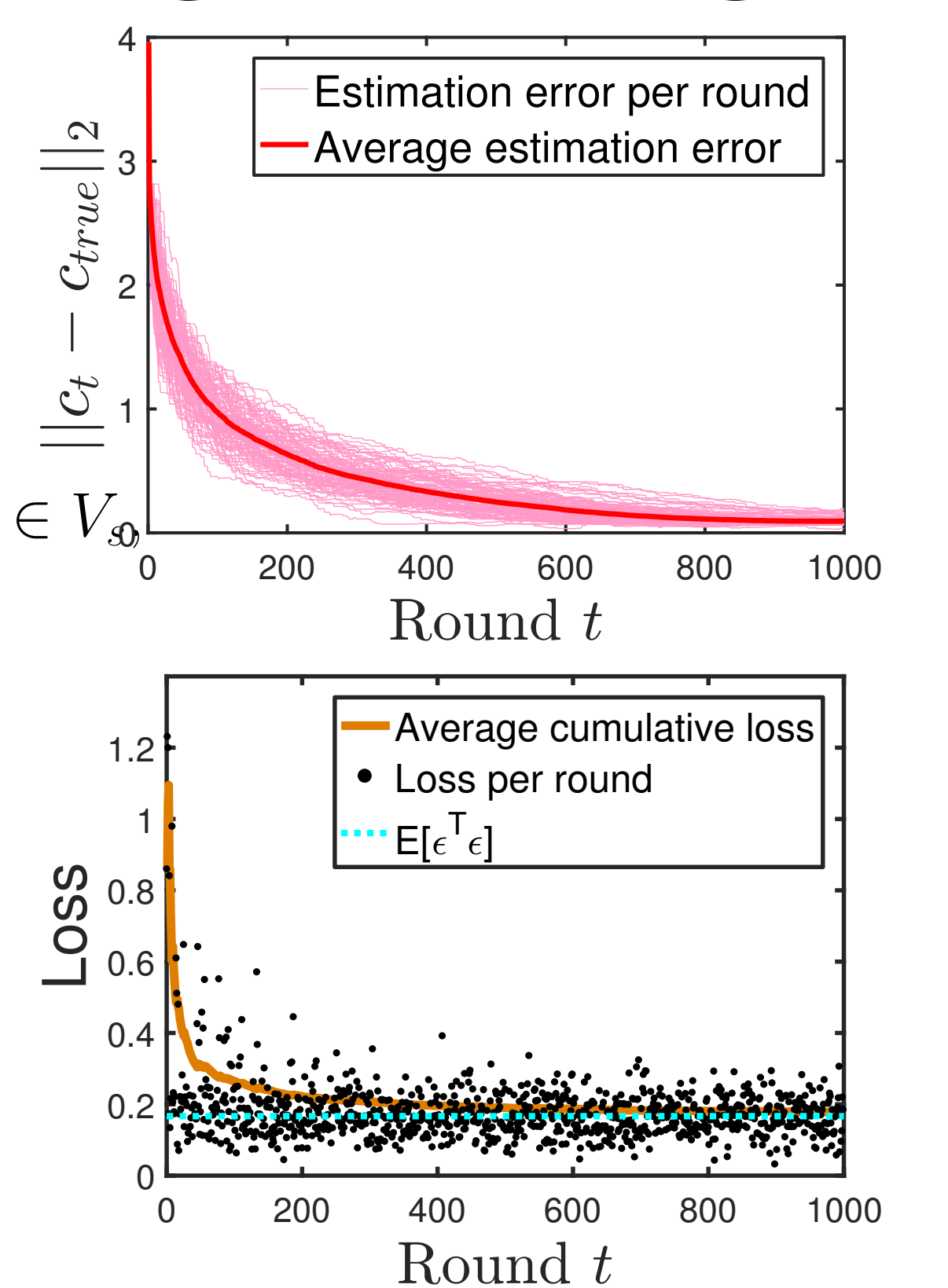
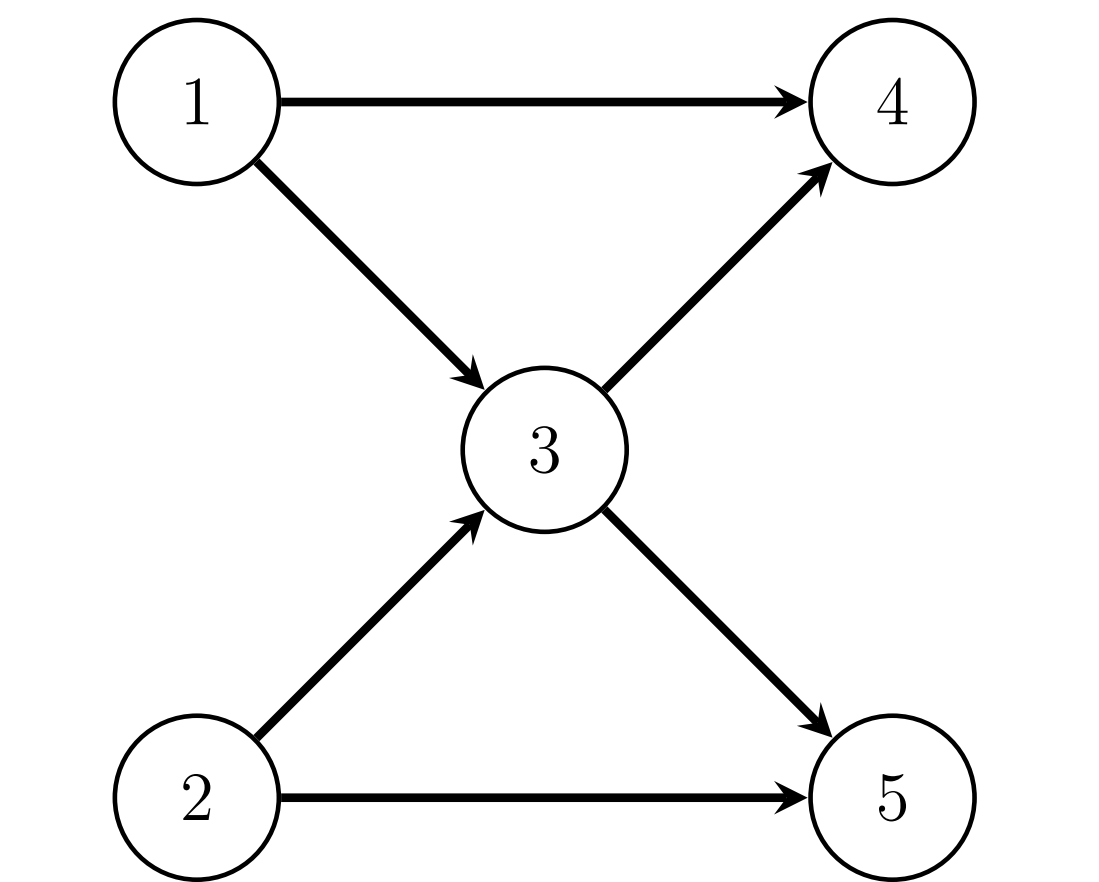
- In this application, we will consider a concave quadratic representation for  $u(\mathbf{x})$ . That is,  $u(\mathbf{x}) = \frac{1}{2} \mathbf{x}^T Q \mathbf{x} + \mathbf{r}^T \mathbf{x}$ , where  $Q \in \mathbb{S}_-^n$  (the set of symmetric negative semidefinite matrices),  $\mathbf{r} \in \mathbb{R}^n$ .



## Learning transportation cost

- We now consider the transshipment network  $G = (V_s \cup V_d, E)$ , where nodes  $V_s$  are producers and the remaining nodes  $V_d$  are consumers.
- The transshipment problem can be formulated in the following:

$$\begin{aligned} \min \quad & \sum_{v \in V_s} C^v(y_v) + \sum_{e \in E} c_e x_e \\ \text{s.t.} \quad & \sum_{e \in \delta^+(v)} x_e - \sum_{e \in \delta^-(v)} x_e = y_v, \quad \forall v \in V_s, \\ & \sum_{e \in \delta^+(v)} x_e - \sum_{e \in \delta^-(v)} x_e = d_v^l, \quad \forall v \in V_d, \\ & 0 \leq x_e \leq u_e, \quad 0 \leq y_v \leq w_v, \quad \forall e \in E, \forall v \in V_s. \end{aligned}$$



where we want to learn the transportation cost  $c_e$  for  $e \in E$ . For this application, we will consider a convex quadratic cost for  $C^v(y_v)$ . That is,  $C^v(y_v) = \frac{1}{2} \lambda_v y_v^2$ , where  $\lambda_v \geq 0$ .

## Conclusions

- We first propose the general framework for eliciting decision maker's preferences and restrictions using inverse optimization through online learning.
- This framework can learn general convex utility functions and restrictions with observed noisy signal-decision pairs.
- Moreover, we prove that the online learning algorithm, which uses an implicit update rule, has a  $\mathcal{O}(\sqrt{T})$  regret under certain regularity conditions. Hence, this method has a fast convergence rate.

Code at <https://github.com/chaoshengdong/GIOPOL>  
 Paper at <https://arxiv.org/pdf/1810.01920.pdf>

